



# E3 ligase BRG3 persulfidation delays tomato ripening by reducing ubiquitination of the repressor WRKY71

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## Abstract

Hydrogen sulfide (H<sub>2</sub>S) is a gaseous signaling molecule reported to play multiple roles in fruit ripening. However, the molecular mechanisms underlying H<sub>2</sub>S-mediated delay in fruit ripening remain to be established. Here, the gene encoding a WRKY transcription factor, *WRKY71*, was identified as substantially upregulated in H<sub>2</sub>S-treated tomato (*Solanum lycopersicum*) via transcriptome profiling. The expression of *WRKY71* was negatively associated with that of *CYANOALANINE SYNTHASE1* (*CAS1*). Transient and stable genetic modification experiments disclosed that *WRKY71* acts as a repressor of the tomato ripening process. *CAS1* appears to play an opposite role, based on the finding that the ripening process was delayed in the *cas1* mutant and accelerated in *CAS1*-OE tomatoes. Dual-luciferase reporter assay, yeast one-hybrid, electrophoretic mobility shift assay, and transient transformation experiments showed that *WRKY71* bound to the *CAS1* promoter and suppressed its activation. Moreover, the persulfidation of *WRKY71* enhanced its binding ability to the *CAS1* promoter. Data from luciferase complementation and Y2H assays confirmed that *WRKY71* interacts with a BOI-related E3 ubiquitin-protein ligase 3 (*BRG3*) and is ubiquitinated *in vitro*. Further experiments showed that modification of *BRG3* via persulfidation at Cys<sub>206</sub> and Cys<sub>212</sub> led to reduced ubiquitination activity. Our findings support a model whereby *BRG3* undergoes persulfidation at Cys<sub>206</sub> and Cys<sub>212</sub>, leading to reduced ubiquitination activity and decreased interactions with the *WRKY71* transcript, with a subsequent increase in binding activity of the persulfidated *WRKY71* to the *CAS1* promoter, resulting in its transcriptional inhibition and thereby delayed ripening of tomatoes. Our collective findings provide insights into a mechanism of H<sub>2</sub>S-mediated regulation of tomato fruit ripening.

## Introduction

Tomato (*Solanum lycopersicum*) is a classical climacteric fruit extensively used as a model to study fruit ripening and senescence. Multiple physiological and biochemical reactions occur during ripening and senescence of tomato fruits, including pigment metabolism and transformation as well as synthesis and degradation of plant hormones (Giovannoni et al. 2016), which are regulated by a series of transcription factors (TFs). WRKY, a unique TF family in plants, participates in plant signal transmission, ethylene biosynthesis, abiotic stress responses, and the process of ripening and

senescence (Rushton et al. 2010). Generally, WRKY domains contain 1 or 2 highly conserved amino acid sequences (WRKYGQK) and a zinc-finger motif that binds the (T)(T)TGCA(C/T) element (W-box) of downstream gene promoters. In *Arabidopsis* (*Arabidopsis thaliana*), *WRKY53* is reported to regulate leaf senescence through binding the W-box and activating the expression of *Senescence associated gene 12* (*SAG12*), *CATALASE1/2/3* (*CAT1/2/3*), and *ORESARA 9* (*ORE9*) (Schippers 2015). In tomatoes, *WRKY35* enhances carotenoid accumulation by directly activating *DXS1*, *WRKY31*, and *WRKY23* at the breaker and red-ripe maturity

stages (Bartley and Ishida 2003; Yuan et al. 2022). In peach (*Prunus persica*), WRKY22 ubiquitination by ring-type E3 ligase HOS1 is proposed to play a role in regulating the defense program (Li et al. 2020a, 2020b). A similar regulatory mode of WRKY45 has been reported, whereby its ubiquitination by nuclear ubiquitin-proteasome modulates the plant defense program in rice (Akane et al. 2013). The collective findings to date suggest that WRKYs are involved in various physiological regulation mechanisms in plants through direct binding to target genes or posttranslational modifications. However, the issue of whether WRKY TFs can exert regulatory effects on tomato fruit ripening through ubiquitination remains unclear.

During fruit ripening, hydrogen cyanide (HCN), CO<sub>2</sub>, and ethylene are concomitantly generated (Floss et al. 1965). Cyanoalanine synthase (CAS) is an enzyme that converts HCN to cyanoalanine and H<sub>2</sub>S via the decyanidation pathway in plants (Warrilow 1998). García et al. (2014) reported increased cyanogen content in roots, thereby inhibiting the formation of root hairs in plants with mitochondrial  $\beta$ -cyanoalanine synthase CYS-C1 deletion in *Arabidopsis*. In avocado and carnation flowers, CAS plays a substantial role in decyanation. However, as CAS activity increases with the production of a large amount of ethylene, the HCN level is almost undetectable (Manning 1986; Yip and Yang 1998). Recent reports have disclosed markedly higher CAS activity in climacteric than non-climacteric fruits (Mizutani et al. 1988). In apples, CAS1 and CAS2 potentially play a role in cyanide detoxification during ripening (Sang et al. 2007). However, no direct evidence of a regulatory role of CAS in the ripening process in fruits has been obtained to date.

H<sub>2</sub>S is a gaseous signaling molecule with important physiological functions in root formation, flower opening, stress response, fruit ripening, and senescence (He et al. 2018; Zhang et al. 2020; Huang et al. 2021). Exogenous H<sub>2</sub>S suppresses respiration and ethylene production in bananas, strawberries, kiwi, and other fruits (Hu et al. 2012; Li et al. 2017; Ge et al. 2018). Upon knock-out of the endogenous H<sub>2</sub>S production enzyme LCD1, the tomato fruit ripening process is accelerated (Hu et al. 2020), indicating a regulatory role of H<sub>2</sub>S in fruit ripening. However, the underlying molecular mechanisms remain to be established. Accumulating evidence has shown that ~5% proteins are persulfidated in *Arabidopsis*, a posttranslational modification whereby the thiol group at cysteine residues (–SH) is replaced with the persulfidation group (–SSH). The majority of persulfidated proteins are implicated in the nuclear localization signal (Khan et al. 2018), signal transduction (Chen et al. 2020), cell autophagy (Ma et al. 2021), and flowering processes (Laureano et al. 2020). Due to the different nucleophilic abilities of the –SH and –SSH groups, modification by persulfidation confers stronger chemical reactivity (Paul and Snyder 2012; Corpas 2019). For instance, persulfidation of Cys<sub>160</sub> of glyceraldehyde-3-phosphate dehydrogenase by H<sub>2</sub>S enhances its nuclear localization (Khan et al. 2018). In guard cells, H<sub>2</sub>S-induced persulfidation of Cys<sub>131</sub> and Cys<sub>137</sub> sites of OST1/SNF1-related

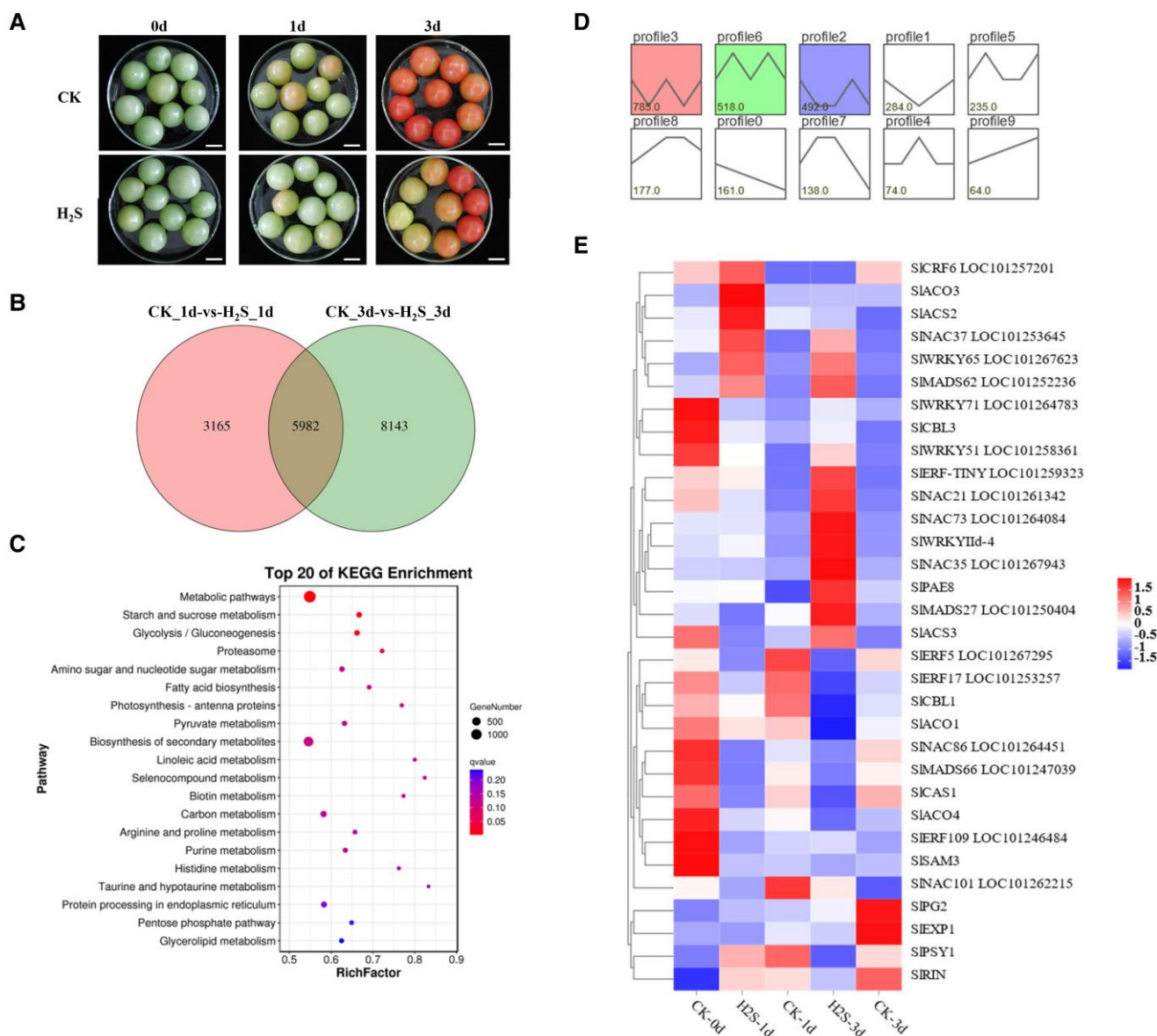
protein kinase 2.6 (SnRK2.6) enhances kinase activity, thereby promoting the closure of stomata (Chen et al. 2020). Additionally, H<sub>2</sub>S is reported to inhibit the binding of FLC1 and FLC3 to promoters of target genes through persulfidation, thereby inducing the flowering process in Chinese cabbage (Ma et al. 2021). While H<sub>2</sub>S-mediated persulfidation is clearly involved in several aspects of plant development, its precise role in the regulation of tomato fruit ripening is yet to be established.

To explore the mechanisms underlying H<sub>2</sub>S regulation of the tomato ripening process, we screened the TF WRKY71 responding to H<sub>2</sub>S signal from transcriptome data of H<sub>2</sub>S-treated tomato fruit, which showed a negative association with cyanoalanine synthase CAS1. Moreover, WRKY71 could interact with BRG3 and both proteins were modified by persulfidation. Based on the data obtained, we proposed that H<sub>2</sub>S-mediated persulfidation of BRG3 and WRKY71 could weaken BRG3's ubiquitination ability and enhance the binding ability of WRKY71 to CAS1, thereby delaying ripening of tomato fruit. To validate this hypothesis, we investigated the functions of WRKY71 and CAS1 in tomato fruit by transient expression and stable genetic system. The regulatory relationship between WRKY71 and CAS1 was identified by a dual-luciferase reporter, yeast one-hybrid (Y1H), electrophoretic mobility shift assay (EMSA), and transient transformation assays. Moreover, we identified persulfidation of BRG3 weakened its ubiquitination ability and the interaction with WRKY71 by in vitro simulation experiments, yeast two-hybrid (Y2H), and luciferase complementation analysis. Data from the current study provide insights into the molecular mechanisms underlying H<sub>2</sub>S-mediated regulation of tomato fruit ripening and posttranslational modification of proteins.

## Results

### WRKY71 screened from the transcriptome shows a negative association with CAS1 in H<sub>2</sub>S-treated tomato fruits

In a recent study, Zhong et al. (2021) showed that the ripening process of tomato fruit could be delayed by exogenous H<sub>2</sub>S treatment. Tomatoes did not turn completely red at 3 d under H<sub>2</sub>S treatment while the color changed entirely in tomatoes subjected to deionized water treatment (CK) (Fig. 1A). To establish the molecular mechanisms underlying the H<sub>2</sub>S-mediated delay in fruit ripening, we performed RNA-seq of CK\_0d, CK\_1d, H<sub>2</sub>S\_1d, CK\_3d, and H<sub>2</sub>S\_3d samples. The number of clean-read libraries was 591–686 million (Supplemental Table S1). Based on statistical alignment data, the alignment efficiency of clear reads to the reference genome ranged from 88.91% to 92.17% (Supplemental Table S1). Pairwise comparison of CK\_1d vs H<sub>2</sub>S\_1d and CK\_3d vs H<sub>2</sub>S\_3d led to the identification of a total of 5,982 shared differentially expressed genes (DEGs) (Fig. 1B). Analysis of these DEG profiles revealed 10 major expression patterns, among

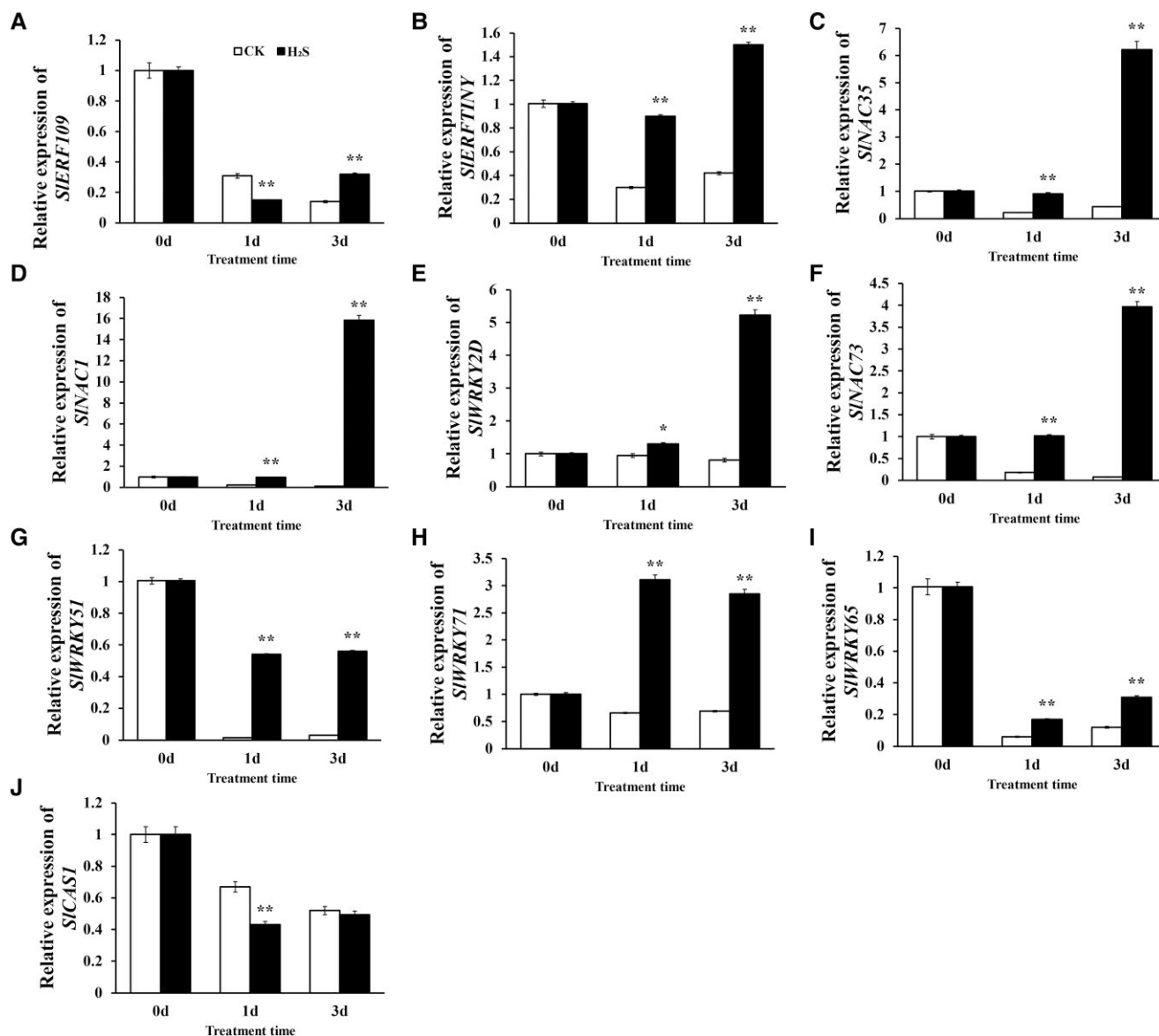


**Figure 1** Transcriptomic analysis of tomato fruit treated with H<sub>2</sub>S or control (CK) at 0, 1, and 3 d. A) Phenotype analysis of tomato fruit treated with H<sub>2</sub>S for 0, 1, and 3 d at the BR stage. Scale bars = 1 cm. B) Screening of DEG between groups of CK\_1d vs H<sub>2</sub>S\_1d and CK\_3d vs H<sub>2</sub>S\_3d. C) KEGG analysis of common DEGs screened from groups of CK\_1d vs H<sub>2</sub>S\_1d and CK\_3d vs H<sub>2</sub>S\_3d. D) Expression pattern analysis of DEGs between groups of CK\_1d vs H<sub>2</sub>S\_1d and CK\_3d vs H<sub>2</sub>S\_3d. E) Expression patterns of ripening-related genes identified from common DEGs. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with Student's  $t$ -test, and significance was marked with asterisks: \* ( $P < 0.05$ ) or \*\* ( $P < 0.01$ ).

which profiles 3, 4, and 9 were upregulated DEGs from H<sub>2</sub>S-treated tomato fruits (Fig. 1D). These DEGs were involved in numerous essential processes, such as carbon metabolism, abiotic and biological stress responses, growth and development, and signal transduction, as determined from Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis (Fig. 1C). CAS1, CBL1, CBL3, ACO1, ACO3, ACO4, ACS2, ACS3, SAM3, PG2, PSY1, EXP1, and PAE8 were identified as DEGs in H<sub>2</sub>S-treated compared with CK-treated tomato fruits (Fig. 1E), suggestive of potential involvement in the regulatory processes of tomato fruit ripening. The majority of these genes are known participants in fruit ripening. The

function of CAS1 (LOC101253940) in fruit ripening has not been established so far and this gene was, therefore, selected as a candidate for further study.

A total of 18 differentially expressed TFs were screened, including 4 WRKYs, 3 MADSs, 5 ERFs, and 6 NACs. Expression levels of WRKY2D, WRKY71, WRKY51, WRKY65, ERF109, ERF109, ERF109, NAC73, NAC1, NAC35, and CAS1 were further analyzed via RT-qPCR. The results showed the upregulation of ERF109, ERF109, NAC73, NAC1, NAC35, WRKY2D, WRKY71, WRKY51, and WRKY65 in H<sub>2</sub>S-treated compared with CK-treated groups, conversely, the downregulation of CAS1 (Fig. 2, A–J). Correlation analysis showed that CAS1 was



**Figure 2** RT-qPCR analysis of expression levels of candidate genes from tomato fruit treated with H<sub>2</sub>S. Gene expression of A) *ERFB109*, B) *ERFTINY*, C) *NAC35*, D) *NAC1*, E) *WRKY2D*, F) *NAC73*, G) *WRKY51*, H) *WRKY71*, I) *WRKY65*, and J) *CAS1* in tomato fruit treated with H<sub>2</sub>S or CK for 0, 1, and 3 d. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks: \* ( $P < 0.05$ ) or \*\* ( $P < 0.01$ ).

substantially positively correlated with *WRKY65*, *WRKY51*, and *ERFB109* and negatively correlated with *WRKY71*, *WRKY2D*, *NAC1*, *NAC35*, and *ERFTINY* (Fig. 3A; Supplemental Table S2).

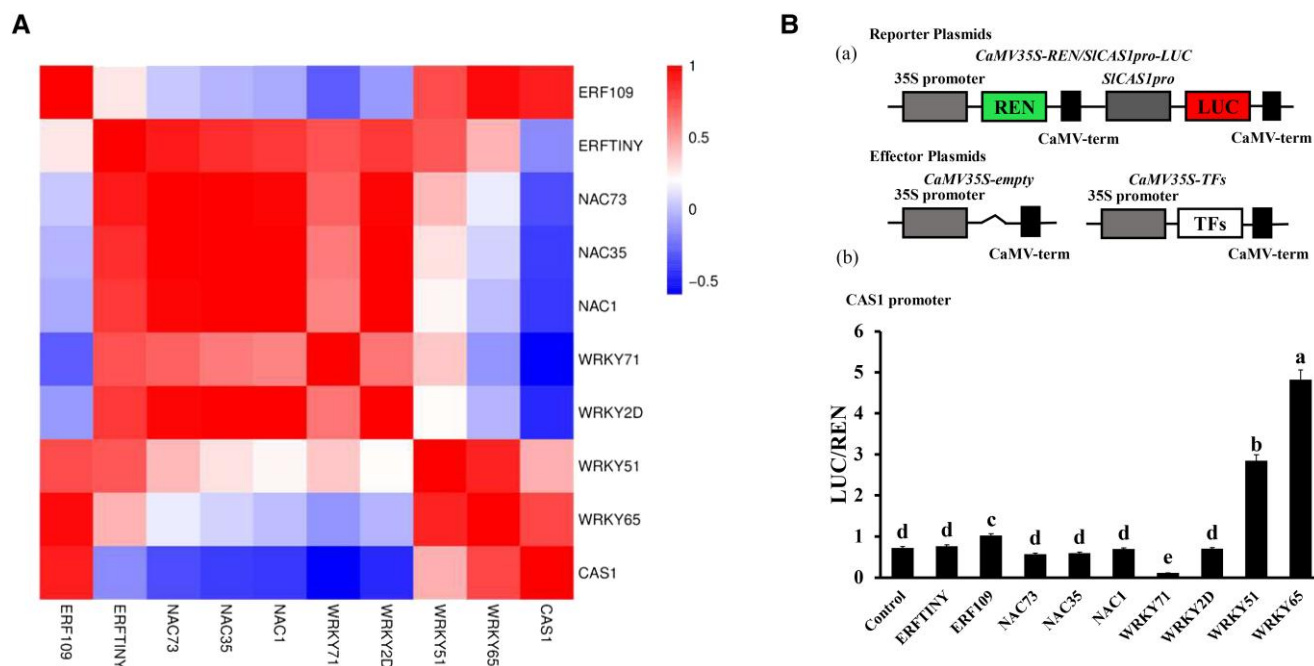
To further explore the transactivation of TFs, we conducted a dual-luciferase reporter assay of *CAS1* in *Nicotiana benthamiana* leaves. As shown in Fig. 3B, *ERFB109*, *WRKY51*, and *WRKY65* significantly activated *CAS1* by 3–5-fold compared with the empty vector ( $P < 0.05$ ). Conversely, *WRKY71* suppressed the activity of the LUC reporter gene by one-seventh relative to the control group ( $P < 0.05$ ), suggesting that *WRKY71* binds to the *CAS1* promoter and inhibits its transactivation. Accordingly, *WRKY71* with the C<sub>2</sub>H<sub>2</sub>

zinc-finger domain was selected for functional analysis in tomato fruit ripening (Supplemental Fig. S1).

### *WRKY71* acts as a repressor in the regulation of tomato fruit ripening

To establish the function of *WRKY71* in the ripening process, we performed transient overexpression and virus-induced gene silencing (VIGS) assays. Overexpression of *WRKY71* delayed tomato fruit ripening along with a decreased rate of chlorophyll degradation and carotenoid and lycopene biosynthesis (Supplemental Fig. S2). Compared with the empty vector, the tomato ripening process of TRV-*WRKY71* was accelerated





**Figure 3** Identification of key TFs regulating *CAS1* transcription in tomato fruit. A) Correlation analysis of *ERF109*, *ERFTINY*, *NAC35*, *NAC1*, *WRKY2D*, *NAC73*, *WRKY51*, *WRKY71*, *WRKY65*, and *CAS1* expression. B) Transactivation activity of TFs for the *CAS1* promoter. (a) Model of reporter and effector plasmids. (b) TFs activating the *CAS1* promoter were validated with the dual-luciferase reporter assay. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with One-way ANOVA, and different letters signify  $P < 0.05$ .

and chlorophyll degradation, carotenoid biosynthesis, and lycopene biosynthesis enhanced (Supplemental Fig. S3).

We further obtained stable genetic *wrky71* mutant plants via CRISPR/Cas9 technology and overexpression (*WRKY71*-OE) transgenic plants (Supplemental Fig. S4). Phenotypic analysis of tomato fruits of *wrky71*, WT, and *WRKY71*-OE samples was shown in Fig. 4A. The *wrky71* mutant tomato fruits displayed the break red (BR) stage at 28 DPA and became mature red (MR) at 37 DPA, whereas WT tomatoes entered BR at 37 DPA and MR at 46 DPA. *WRKY71*-OE tomato entered the mature white stage at 36 DPA and eventually entered the mature red stage at 50 DPA. Overall, *wrky71* tomato fruit matured 9 d earlier and *WRKY71*-OE fruit matured over 13 d later than WT. The chlorophyll content presented a declining trend during the ripening process (Fig. 4C). At 37 DPA, the chlorophyll contents of WT, *WRKY71*-OE, and *wrky71* fruit were 3.1, 2.7, and 1.7 mg/g, respectively. The carotenoid content in the *wrky71* sample increased during the ripening process and was higher than that of WT at 37 DPA, but decreased at 46 DPA to a lower extent than WT (Fig. 4D). The carotenoid content in *WRKY71*-OE fruit was higher than in both *wrky71* and WT at 47 DPA ( $P < 0.05$ ). The lycopene content in *wrky71*, *WRKY71*-OE, and WT tomato plants increased during the ripening process and was consistently higher in the *wrky71* mutant than in *WRKY71*-OE and WT tomato ( $P < 0.01$ ) (Fig. 4E).

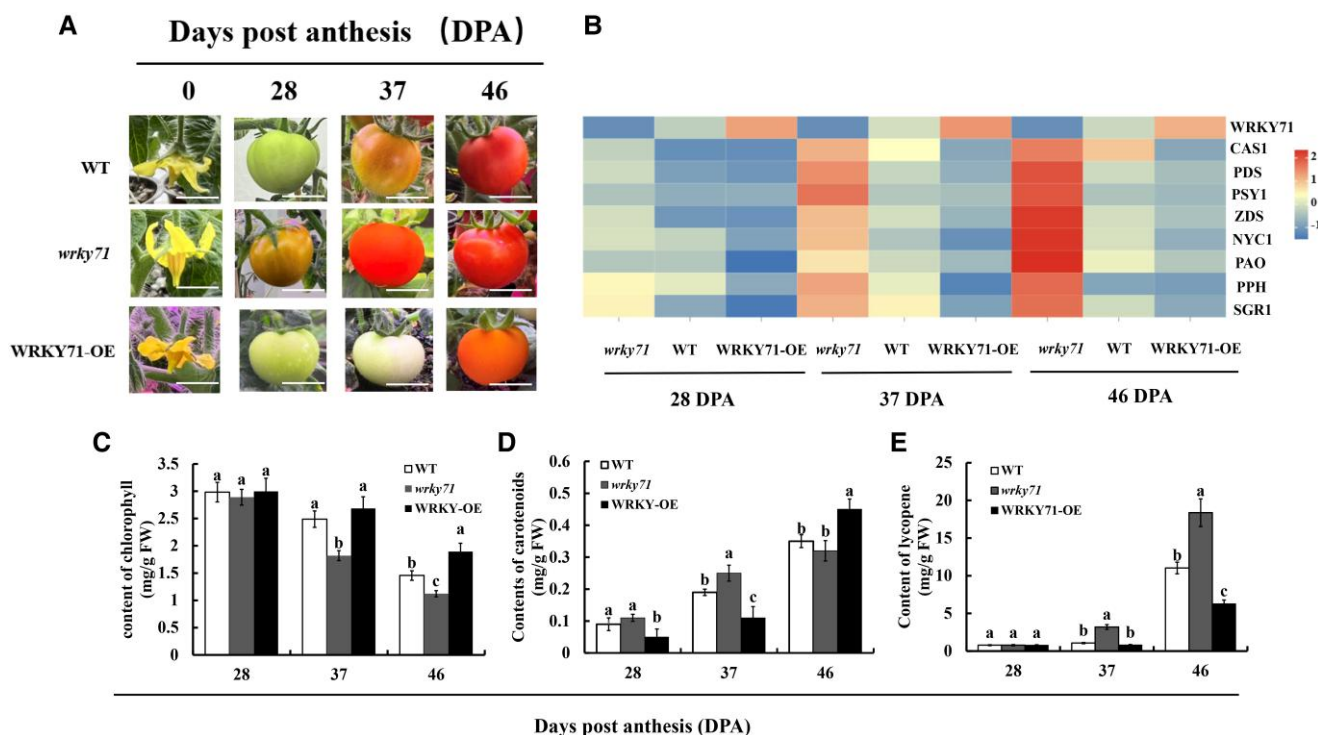
We analyzed the expression patterns of key genes involved in the chlorophyll degradation pathway in *wrky71* mutant, *WRKY71*-OE transgene, and WT tomato fruits, including

*NYC1*, *PAO*, *PPH*, and *SGR1*. As shown in Fig. 4B, the expression of all 4 genes was substantially increased at 28 DPA in the *wrky71* mutant up to a more than 10-fold higher extent relative to WT, while the relative levels of these genes in *WRKY71*-OE were lower than WT. Expression trends of the key genes in carotenoid biosynthesis were similar to those of the chlorophyll degradation pathway genes in all 3 tomato fruit groups. Expression levels of *PDS*, *PSY1*, and *ZDS* in *wrky71* fruits were 10–70-fold higher relative to WT at 28 DPA. However, expression levels of these genes were increased at 37 DPA in WT and *WRKY71*-OE fruits. Moreover, the expression of *CAS1* in the *wrky71* mutant tomato fruit consistently remained the highest during the ripening process while that in *WRKY71*-OE was the lowest (Fig. 4B).

### Determination of the function of *CAS1* in tomato fruit ripening

The functional role of *CAS1* in tomato fruit was explored using VIGS. Tomato fruits with *CAS1* silencing were explored at the late-maturing stage. Notably, chlorophyll degradation, carotenoid biosynthesis, and lycopene biosynthesis were decelerated compared with control (Supplemental Fig. S5).

We obtained genetically stable *CAS1* mutant plants via CRISPR/Cas9 technology. And we obtained transgenic tomato that was generated by 35S:*CAS1*, designated *CAS1*-OE (Supplemental Fig. S6). The expression of *CAS1* was substantially inhibited in the *CAS1* mutant during the tomato ripening process. Conversely, the expression of *CAS1* in *CAS1*-OE was nearly 20 times higher than that in WT tomato fruit



**Figure 4** Functional analysis of *WRKY71* in tomato fruit using CRISPR/Cas9. A) Phenotypic analysis of WT, *wrky71* and *WRKY71*-OE samples at the critical period of maturity. B) Expression patterns of *WRKY71*, *CAS1*, *NYC1*, *PAO*, *PPH*, *SGR1*, *PDS*, *PSY1*, and *ZDS* in WT, *wrky71* and *WRKY71*-OE fruits at the critical period of maturity. C–E) Comparison of chlorophyll, carotenoid, and lycopene contents in WT, *wrky71* and *WRKY71*-OE fruits at the critical period of maturity. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with One-way ANOVA, and different letters signify  $P < 0.05$ . Scale bars = 1 cm.

(Fig. 5B). Phenotypic analysis showed that *CAS1*-OE tomato entered MR at 40 DPA while WT entered MR at 50 DPA, suggesting earlier ripening of *CAS1*-OE fruits (Fig. 5A). In contrast, *CAS1* mutant tomato fruit entered BR at 58 DPA, which was 8 d later than WT (Fig. 5A). Our collective findings indicate that *CAS1* acts as a positive regulator in the tomato fruit ripening process.

The chlorophyll content decreased gradually during the ripening process of WT, *CAS1*-OE, and *cas1* groups, and was consistently the highest in the *cas1* mutant and lowest in *CAS1*-OE tomato fruits (Fig. 5C). Specifically, the chlorophyll content of the *cas1* mutant was 1.3-fold higher than that of WT at 30 DPA ( $P < 0.05$ ) and 1.2-fold higher than that of *CAS1*-OE tomato ( $P < 0.05$ ). The changes in carotenoid content were converse to the chlorophyll degradation pattern (Fig. 5D). At 40 DPA, the carotenoid content in *CAS1*-OE was 2-fold higher than that in WT ( $P < 0.05$ ) and 7-fold higher than that of *cas1* mutant tomato ( $P < 0.05$ ). The trend of lycopene content was similar to that of chlorophyll in *cas1*, WT, and *CAS1*-OE groups (Fig. 5E). At 50 DPA, the lycopene content of *CAS1*-OE was 1.6-fold that of WT ( $P < 0.05$ ) and 7.6-fold that of *cas1* mutant tomato fruit ( $P < 0.05$ ).

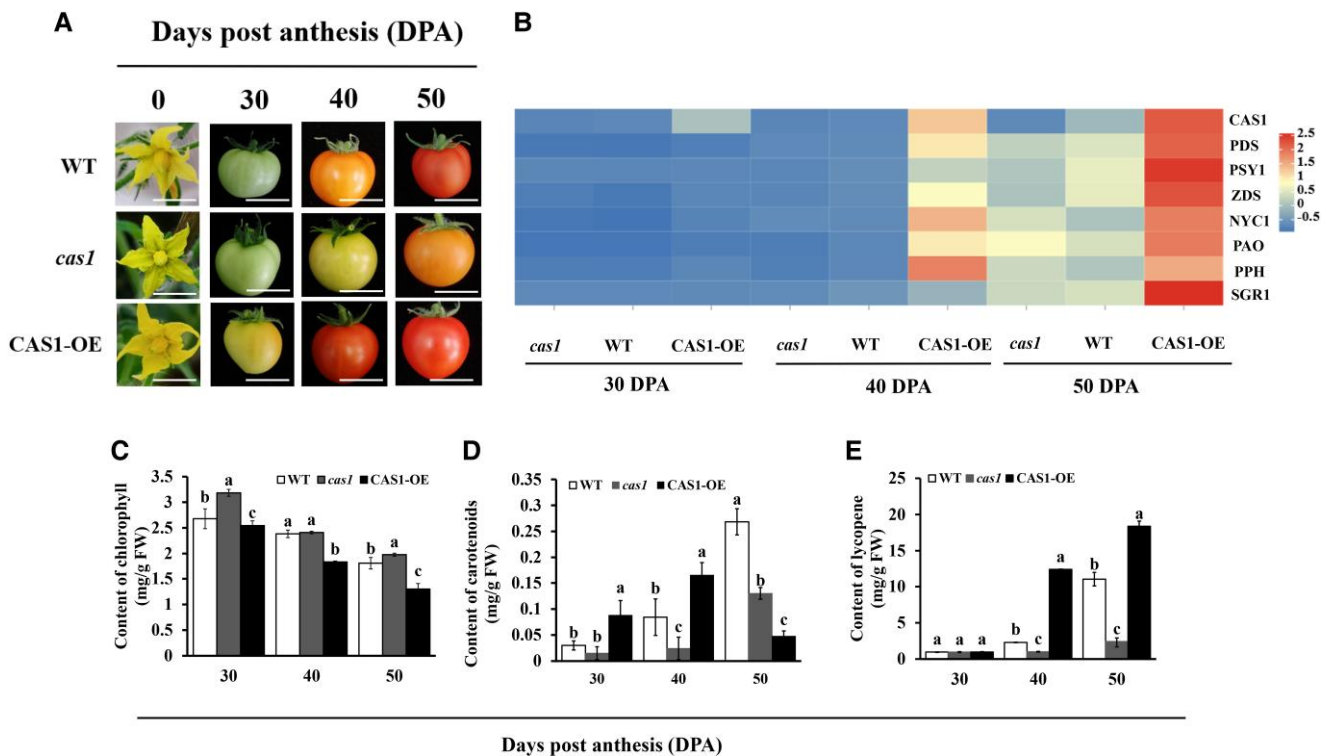
Expression levels of the chlorophyll degradation pathway genes *NYC1*, *PAO*, *PPH*, and *SGR1* in *CAS1*-OE were increased compared with WT, suggesting accelerated chlorophyll degradation in *CAS1*-OE tomatoes. Overall gene levels were

2–3 times higher than those of WT tomatoes at 40 DPA (Fig. 5B). But there were no differences in the expression of these genes which were evident between *cas1* mutant and WT tomato fruits. However, at 50 DPA, *SGR1* expression in the *cas1* mutant was one-fifth that of WT ( $P < 0.05$ ). Expression patterns of *PSY1*, *PDS*, and *ZDS* showed similar trends to *NYC1*, *PAO*, *PPH*, and *SGR1*, with 1-fold higher expression in *CAS1*-OE than WT tomato at 50 DPA ( $P < 0.05$ ) and nearly 2.5-fold higher expression relative to *cas1* mutant fruit ( $P < 0.05$ ) (Fig. 5B).

### *WRKY71* binds the *CAS1* promoter and negatively regulates tomato fruit ripening

To ascertain the regulatory relationship between *CAS1* and *WRKY71*, the activity of *WRKY71* was analyzed using VP16-UAS and the dual-luciferase reporter system. Following truncation of the *WRKY71* coding sequence (CDS) to F1 fragment (0–561 bp) containing the *WRKY*GQK sequence and F2 fragment (520–954 bp) containing the zinc-finger structure, the F2 fragment was identified as the inhibitory region (Fig. 6A).

Analysis of the *CAS1* promoter structure containing multiple *cis*-regulatory elements in PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) revealed the presence of a W-box *cis*-acting element in this



**Figure 5** Functional analysis of CAS1 in tomato fruit via CRISPR/Cas9 and transgene overexpression systems. A) Phenotypic analysis of *cas1*, CAS1-OE, and WT tomato at the critical period of maturity. B) Analysis of relative gene expression of WRKY71, CAS1, NYC1, PAO, PPH, SGR1, PDS, PSY1, and ZDS in *cas1*, CAS1-OE, and WT fruits at the critical period of maturity. C–E) comparison of chlorophyll, carotenoid, and lycopene contents in *cas1*, CAS1-OE, and WT fruits at the critical period of maturity. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with One-way ANOVA, and significance was marked with different letters ( $P < 0.05$ ). Scale bars = 1 cm.

region. The promoter sequence was divided into 3 segments, designated S1 (–1,970 to –1,636 bp), S2 (–1,668 to –1,461 bp), and S3 (–1,197 to –936 bp), and bait vectors (pAbAi-S1, pAbAi-S2, and pAbAi-S3) constructed for the Y1H assay. The results obtained suggest that WRKY71 binds the S1 fragment but not the S2 or S3 of CAS1 (Fig. 6B).

Transient transformation of WRKY71 in CAS1-OE or *cas1* mutant tomato fruits was conducted to establish whether CAS1 is regulated by WRKY71 *in vivo*. As shown in Fig. 6C, overexpression of WRKY71 in CAS1-OE tomatoes delayed the ripening process compared with the group transformed with the empty vector (pSAK277), but no substantial difference in ripening was observed in *cas1* mutant tomato fruits (Fig. 6E). CAS1 expression was repressed upon overexpression of WRKY71 (Fig. 6D), while no differences in CAS1 levels were evident between *cas1* mutant and empty vector groups under conditions of WRKY71 overexpression (Fig. 6F). These data clearly demonstrated that WRKY71-mediated suppression of CAS1 transcription played a role in the regulation of the tomato fruit ripening process.

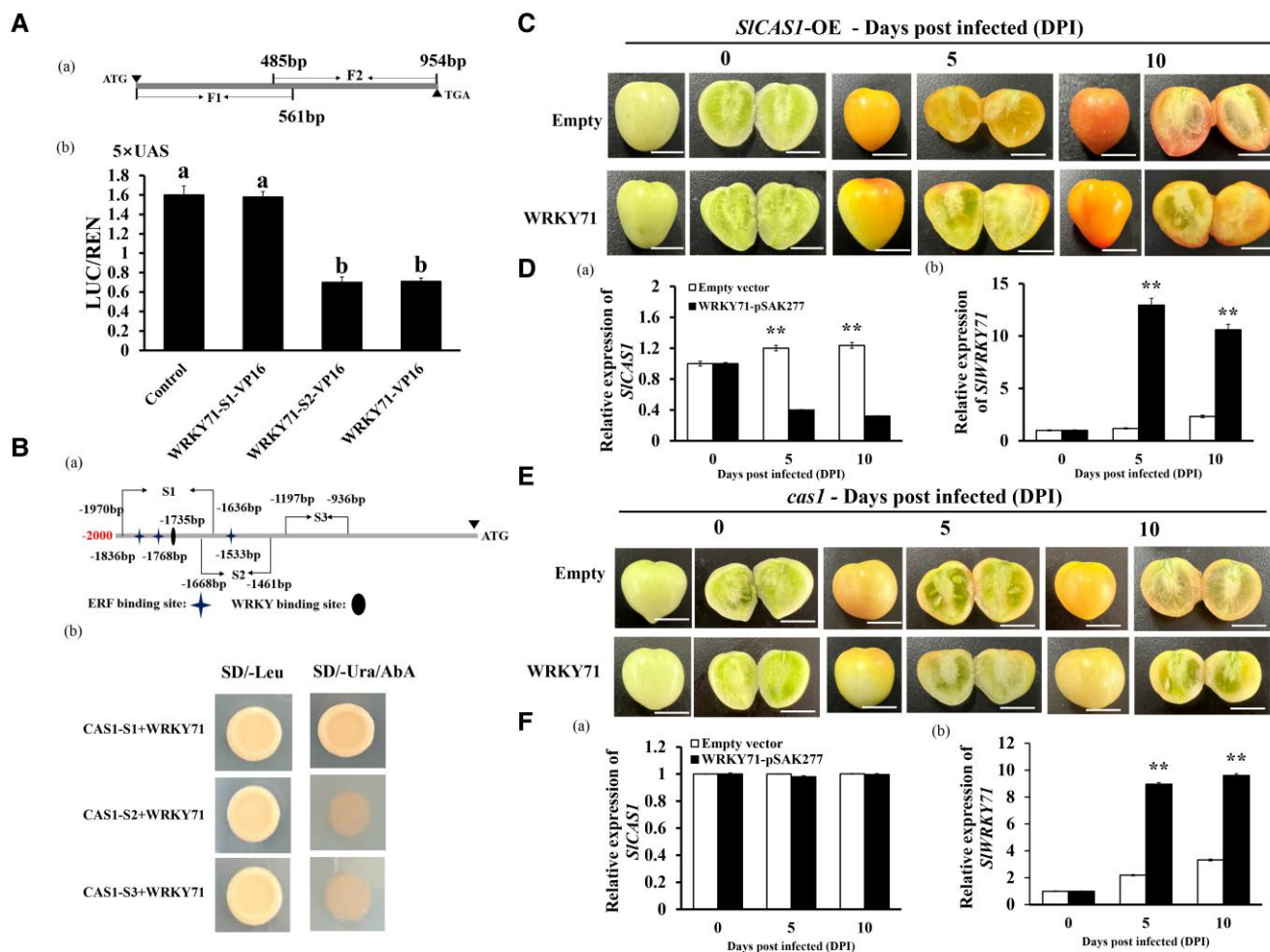
### H<sub>2</sub>S signaling mediates persulfidation and functional regulation of WRKY71

To explore the effect of H<sub>2</sub>S on the regulatory interactions between WRKY71 and CAS1, we conducted the

persulfidation reaction *in vitro* using the biotin switch method. Western blot analysis clearly showed a persulfidation signal in WRKY71 protein treated with H<sub>2</sub>S (Fig. 7A; lanes 2 and 3). In contrast, no band was observed in WRKY71 protein without H<sub>2</sub>S treatment (lane 1) or upon the addition of dithiothreitol (DTT) (lane 4). Additionally, no persulfidation signal was observed in the CAS1 protein (Supplemental Fig. S7). To further validate persulfidation of WRKY71, we constructed a mutation protein in which Cys was replaced with Ala (Fig. 7B) and purified the WRKY71 mutant protein from *Escherichia coli*. As expected, no persulfidation signal was observed for WRKY71 mutant protein treated with H<sub>2</sub>S (Fig. 7B), clearly confirming that the wild-type protein is specifically modified by persulfidation.

To examine the regulatory activity of persulfidated WRKY71 on CAS1, the dual-luciferase experiment was conducted. Distinct from previous experiments, co-transfected *N. benthamiana* leaves were fumigated in H<sub>2</sub>S for 3 h before sampling to determine the function of H<sub>2</sub>S-modified WRKY71. As shown in Fig. 7C, transactivation of WRKY71 under H<sub>2</sub>S fumigation was substantially lower than that without H<sub>2</sub>S treatment ( $P < 0.01$ ). However, we observed no differences in the activation of WRKY71<sup>Cys193AlaCys198Ala</sup> mutants and pSAK277 between groups with and without H<sub>2</sub>S fumigation.





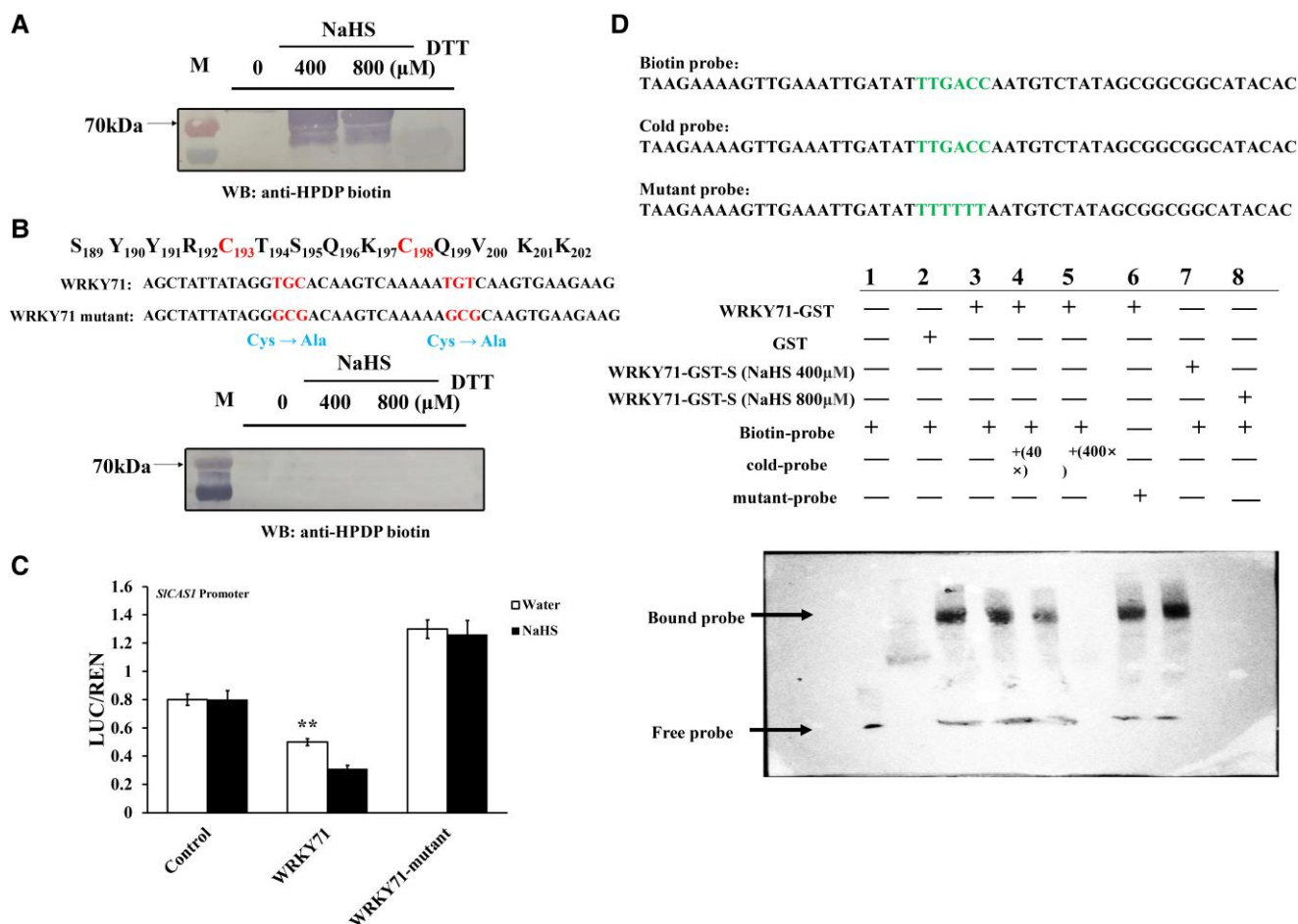
**Figure 6** *WRKY71* suppresses *CAS1* transcription in tomatoes. A) Identification of *WRKY71* as a transcriptional inhibitor. (a) Subfragments of *WRKY71*. (b) Dual-luciferase reporter assay validating the inhibitory activity of *WRKY71* on transcription. B) *WRKY71* directly binds to the *CAS1* promoter. (a) Scheme of the promoter of *CAS1*. Bar segments were analyzed using the PlantCARE database and separated by solid lines (S1, S2, S3). (b) Identification of *WRKY71* bound to the *CAS1* promoter with the Y1H. C) Phenotypic analysis of *CAS1*-OE tomato transfected with *WRKY71* at the critical period of tomato fruit ripening. D) Gene expression analysis of (a) *CAS1* and (b) *WRKY71* in *CAS1*-OE tomato fruit at the critical period of tomato fruit ripening. E) Phenotypic analysis of *cas1* mutant tomato transfected with *WRKY71* at the critical period of tomato fruit ripening. F) Expression of (a) *CAS1* and (b) *WRKY71* in *cas1* mutant tomato transfected with *WRKY71* at the critical period of tomato fruit ripening. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with One-way ANOVA and Student's *t*-test, and significance was marked with asterisks, \* ( $P < 0.05$ ) or \*\* ( $P < 0.01$ ), or different letters ( $P < 0.05$ ). Scale bars = 1 cm.

We additionally performed EMSA experiments using H<sub>2</sub>S-treated *WRKY71* recombinant protein and a W-box-containing *CAS1* promoter probe. As shown in Fig. 7D, bands in lanes 3, 7, and 8 were produced by *WRKY71*-GST protein treated with NaHS (0, 400, and 800  $\mu$ M) and subsequently incubated with biotin-labeled probes. The bands in lanes 7 and 8 were pronounced darker than those in lane 3. These findings indicate that persulfidation increases the ability of *WRKY71* to bind probes. Moreover, upon the addition of cold probes to *WRKY71*-GST protein incubated with biotin probe, weaker bands were detected (lanes 4 and 5). No bands were observed for *WRKY71*-GST incubated with mutant probes (lane 6), suggesting that wild-type *WRKY71* interacts with the W-box in the *CAS1* promoter.

### WRKY71 is ubiquitinated by BRG3 in vitro

According to data from LC-MS/MS analysis, persulfidation occurred in 2 E3 ligases of tomato, RNF144b (LOC101265447) and BRG3 (LOC101267168) (Supplemental Table S3). In view of the previous finding that WRKY TFs interact with E3 ubiquitin ligase to co-regulate physiological processes of plants, we examined the hypothesis that *WRKY71* may be modified via ubiquitination. Accordingly, a *N. benthamiana*-based firefly luciferase complementation assay was conducted to identify interactions of *WRKY71* with RNF-144b or BRG3. The model of vectors is depicted in Fig. 8A. Both BRG3-nLuc+*WRKY71*-cLuc and BRG3-cLuc+*WRKY71*-nLuc showed a stronger capability to rescue luciferase enzyme activity than control constructs (Fig. 8, A and B). Interactions of





**Figure 7** Effects of persulfidation on WRKY71 function. A) Detection of persulfidation of WRKY71 recombinant protein *in vitro*. Purified recombinant WRKY71 protein was treated with the sulfating molecule NaHS (400 or 800 μM) for 30 min and subjected to a modified biotin-switch assay for analysis of protein persulfidation. Treatment with 1 mM DTT was used as a negative control. B) Detection of persulfidation in WRKY71<sup>Cys→Ala</sup> mutant. C) Dual-luciferase reporter assay showing the effect of H<sub>2</sub>S on transactivation of CAS1 transcription by WRKY71. The reporter construct was co-transformed with different versions of WRKY71 in *N. benthamiana* leaf for 48 h. Leaves were treated with 0.1 mM NaHS (or distilled water as a control) for 1 h before sampling. D) EMSA results showing the effect of persulfidation on the binding ability of WRKY71 with CAS1 promoter probes. Data are presented as means ± SD (*n* = 3). Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks: \* (*P* < 0.05) or \*\* (*P* < 0.01).

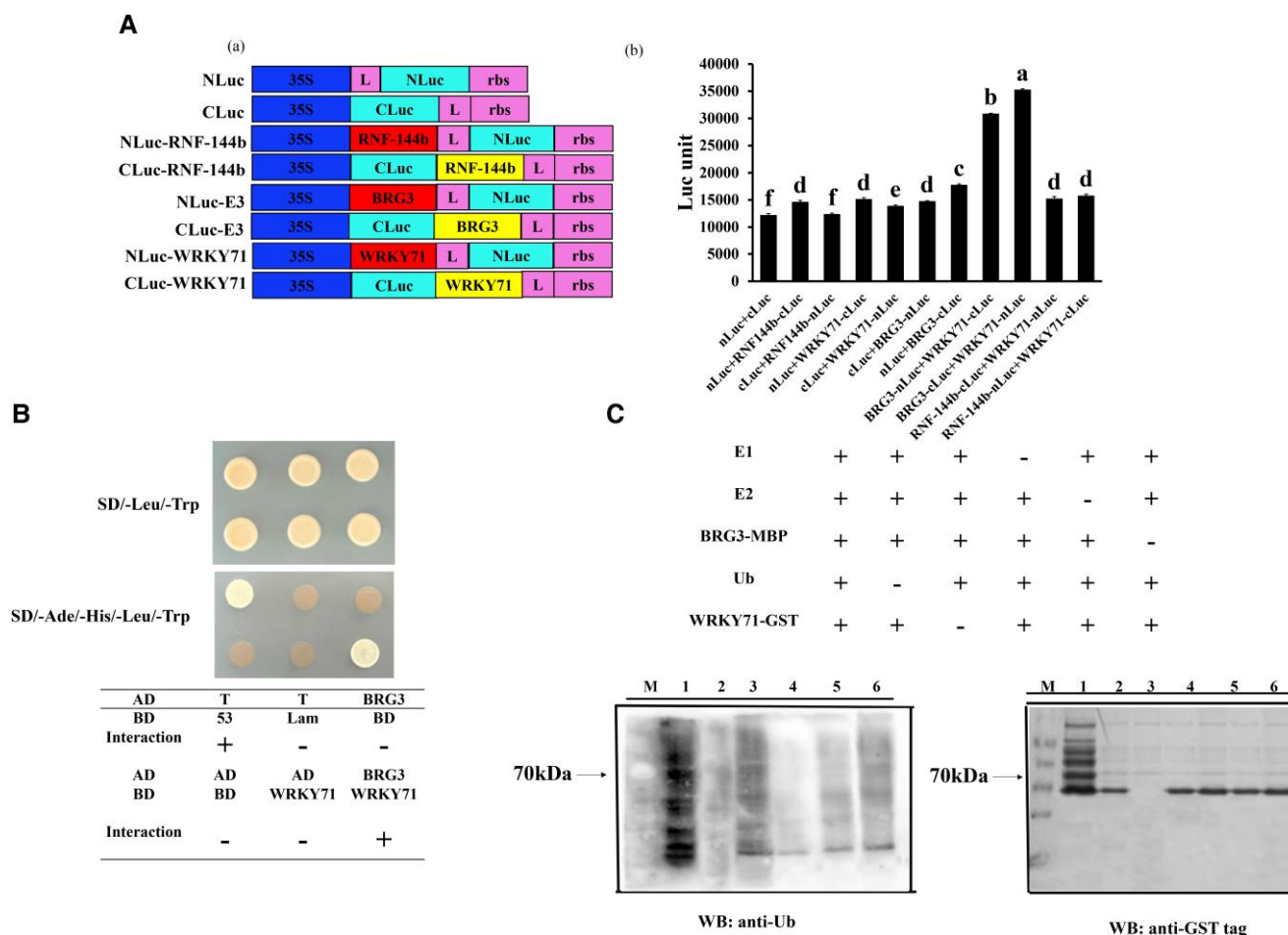
BRG3 with WRKY71 were further verified with the Y2H assay, which showed that Y2H yeast-competent cells containing WRKY71 and BRG3 could grow in medium without Ade, His, Leu, and Trp (Fig. 8B).

To determine whether BRG3 possesses ubiquitin ligase activity, BRG3 protein fused to MBP tag was expressed in *E. coli*, and an *in vitro* ubiquitination assay was conducted. BRG3 displayed self-ubiquitination in human E1 and E2 as well as the substrate Ubs, supporting its activity as a functional ubiquitin ligase. Next, WRKY71 fusion protein with a GST tag was expressed in *E. coli* to establish whether WRKY71 is ubiquitinated by BRG3 protein. As shown in Fig. 8C, polyubiquitinated bands were observed upon incubation of WRKY71 with Ub, E1, E2, and BRG3 proteins in anti-Ub and anti-GST assays (lane 1). In contrast, no polyubiquitinated bands were present in the lanes without Ub, E1, E2, BRG3,

or WRKY71. Our collective findings indicate that BRG3 ubiquitinates WRKY71 *in vitro*.

### Persulfidation and functional regulation of BRG3 mediated by H<sub>2</sub>S signaling

As shown in Fig. 9A, western blot showed a clear persulfidation signal in BRG3 protein treated with H<sub>2</sub>S (lanes 2 and 3), whereas a faint band was observed in BRG3 protein without H<sub>2</sub>S treatment (lane 1) or in the presence of dithiothreitol (DTT) (lane 4). Based on a protein database search, LC-MS/MS analysis revealed the presence of a peptide "SSCMICLPCR" of BRG3 protein, with persulfidation of Cys<sub>3</sub> and Cys<sub>9</sub> (Fig. 9B). Comparison with the protein sequence of BRG3 revealed that Cys<sub>3</sub> and Cys<sub>9</sub> correspond to Cys<sub>206</sub> and Cys<sub>212</sub>, respectively. To validate



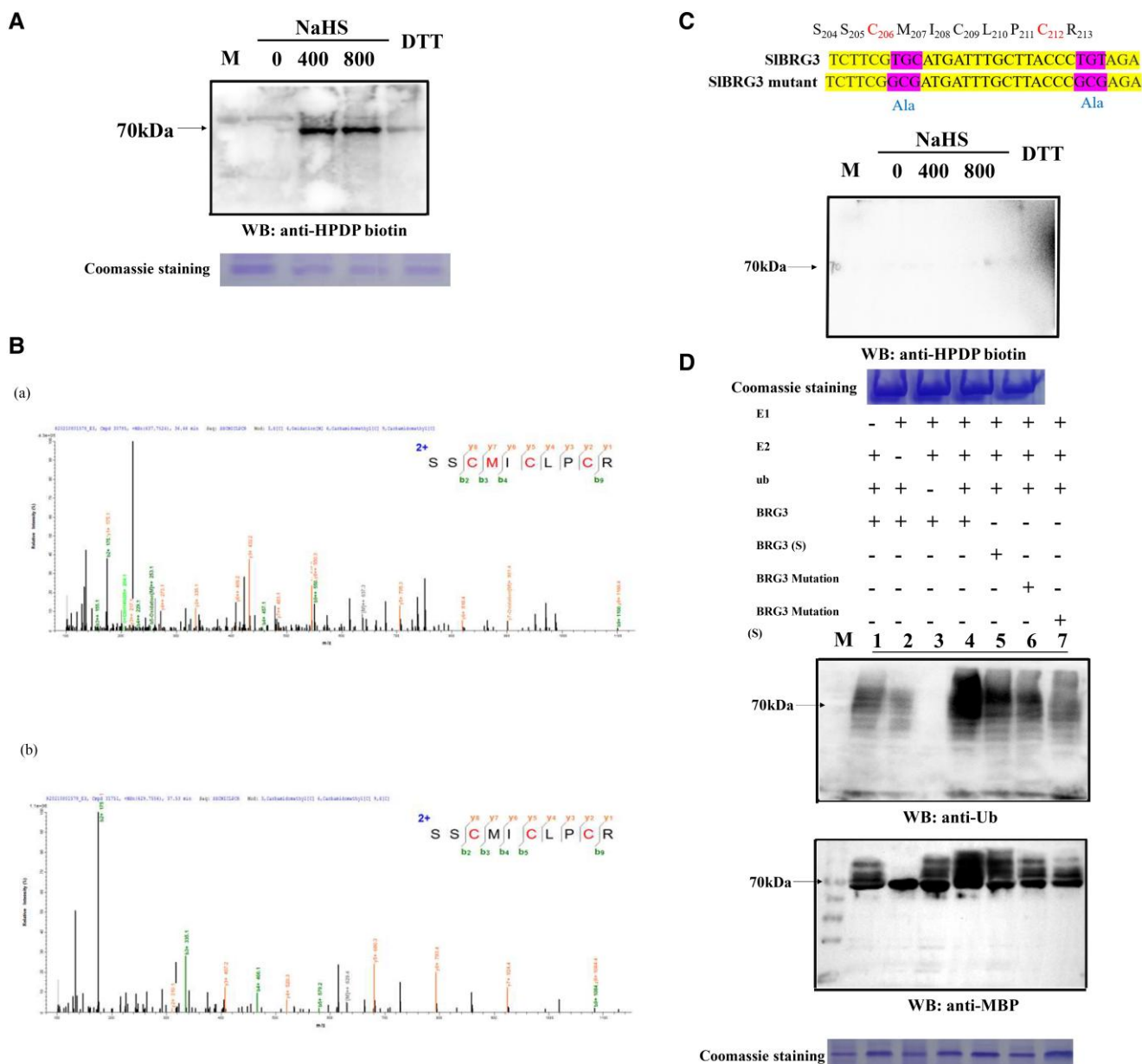
**Figure 8** Identification of BRG3 ubiquitination and interactions with WRKY71. A) Interactions of WRKY71 and BRG3/RNF-144b. (a) Model of nLuc/cLuc, WRKY71-nLuc/cLuc, RNF-144b-nLuc/cLuc, and BRG3-nLuc/cLuc constructs. (b) Firefly luciferase complementation assays of nLuc/cLuc, WRKY71-nLuc/cLuc, RNF-144b-nLuc/cLuc, and BRG3-nLuc/cLuc in 6-week-old *N. benthamiana* leaves. B) Y2H assay of interactions between WRKY71 and BRG3. Positive colonies signify strong interactions between WRKY71 and BRG3. C) Ubiquitination assay validating WRKY71-GST ubiquitination by BRG3-MBP *in vitro*. Data are presented as means  $\pm$  SD ( $n = 3$ ). Statistical analysis was carried out with One-way ANOVA, and significance was marked with different letters ( $P < 0.05$ ).

persulfidation of Cys<sub>206</sub> and Cys<sub>212</sub>, we generated a mutation protein in which Cys residues were replaced with Ala (Fig. 9C) and purified the mutant BRG3 protein from *E. coli*. As expected, no persulfidation signal was observed for BRG3 mutant protein treated with H<sub>2</sub>S (Fig. 9C), strongly suggesting that Cys<sub>206</sub> and Cys<sub>212</sub> are the active cysteine sites modified. An *in vitro* ubiquitination assay identified 4 proteins, BRG3, BRG3-S (protein treated with H<sub>2</sub>S), BRG3<sup>Cys-Ala</sup> mutant, and BRG3<sup>Cys-Ala</sup>-S (protein treated with H<sub>2</sub>S), as functional BRG3 ubiquitin ligases (Fig. 9D). The ubiquitination signal of BRG3-S was weaker compared with BRG3 protein (Fig. 9D; lanes 4 and 5) but similar between BRG3<sup>Cys-Ala</sup> and BRG3<sup>Cys-Ala</sup>-S proteins (Fig. 9D; lanes 6 and 7). There similar results were observed when detected with anti-MBP antibodies and anti-ub antibodies. Our results suggest that persulfidation of BRG3 alters its ubiquitination activity.

## Discussion

### WRKY71 acts as a repressor to negatively regulate tomato fruit ripening

Accumulating evidence has shown that H<sub>2</sub>S delays the fruit ripening process in many horticultural crops (Li et al. 2017; Ge et al. 2018; Zhong et al. 2021). The groups of Yao et al. (2018) and Hu et al. (2019) showed that H<sub>2</sub>S delays tomato fruit ripening by antagonizing the effects of ethylene *in vitro* and *in vivo*. However, the specific molecular mechanisms underlying H<sub>2</sub>S-mediated regulation of tomato fruit ripening are unclear. A number of candidate TF genes responding to H<sub>2</sub>S signalling as potential regulators in fruit ripening were screened via transcriptome analysis, including ERFs, NACs, and WRKYs (Fig. 1). In tomato fruit, among WRKY71, WRKY35, and WRKY23, classified as Group II via genome-wide analysis, WRKY35 and WRKY23 accelerated the tomato ripening process (Bartley and Ishida 2003; Huang et al. 2012;



**Figure 9** Persulfidation of BRG3 at Cys<sub>206</sub> and Cys<sub>212</sub> decreases its ubiquitination activity and interactions with WRKY71. A) Detection of BRG3 recombinant protein persulfidation *in vitro*. Purified recombinant BRG3 protein was treated with the sulfating molecule NaHS (400 or 800  $\mu$ M) for 30 min and subjected to a Modified biotin-switch assay. Treatment with 1 mM DTT was used as a negative control. B) Identification of Cys<sub>3</sub> and Cys<sub>9</sub> corresponding to Cys<sub>206</sub> and Cys<sub>212</sub>, respectively. (a, b) LC-MS/MS analysis of “SSCMICLPCR” from BRG3 protein. C) Detection of persulfidation in BRG3<sup>Cys-Ala</sup>. D) BRG3 persulfidation decreases its ubiquitination ability. M: marker. Ubiquitination abilities of BRG3(S) (lane 5) and mutant BRG3 (lanes 6 and 7) were lower than that of BRG3 (lane 4).

Yuan et al. 2022). Notably, WRKY71 showed a substantial negative correlation with tomato fruit ripening based on RT-qPCR, correlation, and dual-luciferase analysis (Figs. 2 and 3; Supplemental Table S2). Further experiments with a transient system and stable transgenic overexpressing (WRKY71-OE) and mutant plants (*wrky71*) obtained via CRISPR/Cas9 confirmed that WRKY71 functions as a suppressor of tomato fruit ripening (Supplemental Figs. S2 and S3; Fig. 4). The ripening process was accelerated in *wrky71* mutant tomato fruit, pigment contents were increased, and

expression levels of key genes involved in chlorophyll degradation and carotenoid biosynthesis substantially enhanced. The phenotypic characteristics of WRKY71-OE fruits presented an opposite trend (Fig. 4). Our data provide direct evidence that WRKY71 acts as a negative regulator of ripening in tomato fruit. The tomato fruit ripening process is a complex regulatory network involving multiple activators and repressors, such as ripening inhibitor (RIN), non-ripening (NOR), colorless non-ripening (*Cnr*), WRKY35, and WRKY23 (Bartley and Ishida 2003; Ito et al. 2017; Yuan et al. 2022), which act in a



synergistic or antagonistic manner. However, the precise regulatory relationships between *WRKY71* and known ripening regulators require further study.

### **WRKY71 inhibits target gene *CAS1* expression to regulate tomato fruit ripening**

A previous study showed that cyanoalanine synthase 1 (*CAS1*) located downstream of the ethylene production pathway acts as a key rate-limiting enzyme in the decyanidation pathway in plants. *CAS1* and *CAS2* function in cyanide detoxification in apples and are relevant in ethylene production and conversion of HCN in avocado and carnation flowers. Moreover, the activity of *CAS* is higher in climacteric compared with non-climacteric fruits (Manning 1986; Mizutani et al. 1988; Warrilow 1988; Yip and Yang 1998; Sang et al. 2007). These findings suggest that *CAS1* is a key gene involved in the fruit ripening process. In this study, *CAS1* was screened via transcriptome profiling of H<sub>2</sub>S-treated tomato fruit (Fig. 1). Data obtained from a transient expression system, transgenic overexpression (*CAS1*-OE), and *cas1* mutant plants support the role of *CAS1* in promoting tomato ripening. Accordingly, *CAS1* was identified as a positive regulator of ripening (Fig. 5; Supplemental S4). Moreover, *WRKY71* was shown to suppress *CAS1* expression via binding the W-box (Figs. 6, B–D, and 7E). Based on the data, we propose that *WRKY71* acts as a negative regulator of *CAS1* transcription, thereby delaying the tomato ripening process. To clarify the relationship between H<sub>2</sub>S and the tomato fruit ripening mechanism involving the *WRKY71*-*CAS1* axis, persulfidation detection was performed. The results disclosed no signal in *CAS1* while *WRKY71* was modified via persulfidation through the H<sub>2</sub>S donor. Persulfidation increased the ability of *WRKY71* to bind the W-box in the *CAS1* promoter (Fig. 7). These findings were consistent with previous studies showing that persulfidation increases enzyme activity and binding ability in Chinese cabbage (Paul and Snyder 2012; Ma et al. 2021). Accordingly, we conclude that H<sub>2</sub>S activates the transcription of *WRKY71* and induces the persulfidate posttranslational modification of its protein to enhance its binding ability to the *CAS1* promoter, thereby repressing *CAS1* expression and causing a delay in tomato ripening.

### **Persulfidation of BRG3 at Cys<sub>206</sub> and Cys<sub>212</sub> decreases its ubiquitination activity and interactions with WRKY71**

Protein ubiquitination is a common posttranslational modification that participates in the regulation of a wide variety of protein substrates in different cellular pathways through a 3-enzyme cascade (E1–E2–E3) and identification of substrates through Lys residues (Goldknopf et al. 1977; Swatek and Komander 2016). For example, a ring-type E3 ligase, XB3, accelerates the ubiquitination and degradation of NAC2, ACS1, and ACO1 to delay ripening of bananas (Shan et al. 2020) and BRG2/3 ubiquitinates and degrades

a negative regulator, MYB4, thereby accelerating ethylene accumulation to control banana fruit ripening (Yang et al. 2022). The results support a regulatory role of the ubiquitination process in the fruit ripening process. The potential role of persulfidation in ubiquitination may present a mechanism by which H<sub>2</sub>S mediates fruit ripening. In the current study, Y2H and luciferase complementation assays led to the identification of a ring-type E3 ligase, BRG3, that interacts with and ubiquitinates *WRKY71* protein *in vitro* (Fig. 8). It has been shown that BRG3 attenuates the inhibitory effect of *WRKY71* on the fruit ripening process via degradation (Fig. 8), inconsistent with results showing that H<sub>2</sub>S delays the ripening tomato process by inducing an increase in *WRKY71* transcription. Results obtained from LC-MS/MS and mutational analysis in the current study demonstrated that Cys<sub>206</sub> and Cys<sub>212</sub> of BRG3 are modified via persulfidation, with resulting weakening of ubiquitination ability (Fig. 9, B–D). Cys<sub>206</sub> and Cys<sub>212</sub> were thus identified as critical residues for the ubiquitination ability of BRG3. Moreover, BRG3 presents an important target for H<sub>2</sub>S-mediated regulation of physiological processes in tomato fruits. Hence, it is reasonable to speculate that H<sub>2</sub>S signals decrease the ubiquitination activity of BRG3 via persulfidation at Cys<sub>206</sub> and Cys<sub>212</sub>, thereby reducing the degradation of *WRKY71*. Theoretically, the factors underlying changes in protein function caused by persulfidation are not only the different nucleophilic abilities of Cys-SSH but also changes in protein conformation. Further studies are essential to determine whether persulfidation causes changes in protein conformation and the mechanisms by which persulfidation of BRG3 regulates tomato fruit growth and development *in vivo*.

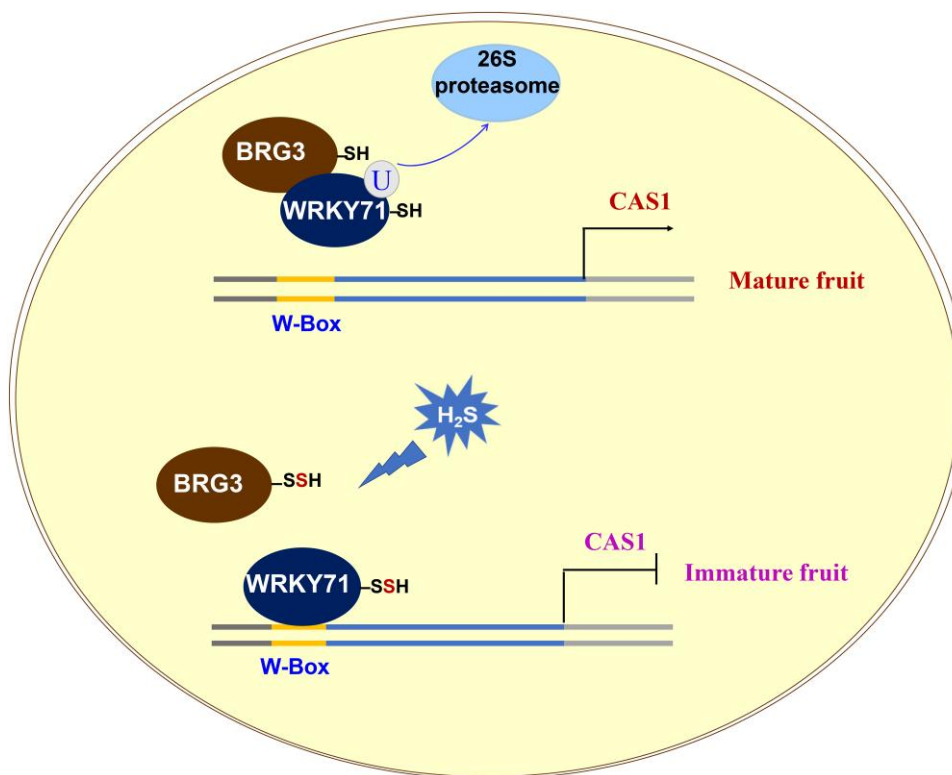
## **Conclusions**

H<sub>2</sub>S is a gaseous signal molecule with multiple roles in fruit ripening and senescence. However, the molecular mechanisms underlying the H<sub>2</sub>S-mediated delay in fruit ripening are currently unclear. In this study, we demonstrated that persulfidation of *WRKY71*, an H<sub>2</sub>S-responsive ripening repressor, increases the binding ability to the *CAS1* promoter and suppresses the transcription of *CAS1*, thereby delaying tomato ripening. Moreover, Cys<sub>206</sub> and Cys<sub>212</sub> of the E3 ubiquitin ligase, BRG3, were modified by persulfidation, which reduced its ubiquitination activity and interactions with *WRKY71*. Our collective findings support a model in which H<sub>2</sub>S persulfidates BRG3<sup>Cys206</sup> and BRG3<sup>Cys212</sup>, resulting in reduced degradation of *WRKY71*. Meanwhile, persulfidation of *WRKY71* promotes transcriptional inhibition of *CAS1*, resulting in a delay in the ripening process of tomato fruit (Fig. 10).

## **Materials and methods**

### **Plant materials and growth**

'Micro Tom' tomatoes (*S. lycopersicum*) used in this study were grown in the glass house of the School of Food and



**Figure 10** Postulated working model of association of WRKY71 with BRG3 to regulate CAS1 expression, thereby controlling tomato fruit ripening. WRKY71 inhibits CAS1 expression by binding to the W-box in the promoter region and BRG3 ubiquitinates and degrades WRKY71. Persulfidation of BRG3 induced by H<sub>2</sub>S signaling leads to a decrease in its ubiquitination activity and consequent WRKY71 degradation and persulfidation of WRKY71 promote inhibition of CAS1 expression, resulting in delayed tomato fruit ripening. Arrow, positive effect; bar, negative regulation; orange rectangle, W-box.

Biological Engineering (Hefei University of Technology, Hefei, China). Tomato plants were grown on nutrient soil: vermiculite (3:1, v/v) in 10 cm pots in an environment-controlled growth room (23 ± 2 °C; 50–70% relative humidity [RH]) under conditions of 16 h light/8 h dark and 250 μmol/m<sup>2</sup>/s light.

The 4-week-old tobacco (*N. benthamiana*) leaves were used for dual-luciferase reporter system and firefly luciferase complementation assays. And the samples were frozen in liquid nitrogen and stored at –80 °C.

### RNA extraction and transcriptome sequencing using the Illumina Platform

According to the method of Zhong et al. (2021), 60 tomato fruits at the stage of color break were selected and divided into 6 groups (10 tomatoes per group) representing 3 replicates of control (CK) (250 ml distilled water) and H<sub>2</sub>S treatment (250 ml, 0.9 mM NaHS aqueous solution) groups, respectively. These tomato samples were selected for transcriptome analysis. Untreated fruit was designated CK\_0d, H<sub>2</sub>S-treated fruits for 1 and 3 d denoted H<sub>2</sub>S\_1d and H<sub>2</sub>S\_3d, and control fruits watered for 1 and 3 d denoted CK\_1d and CK\_3d, respectively. Total RNA was extracted with TRIzol and RNA molecules within a size range of 18–30 nt enriched via polyacrylamide gel electrophoresis (PAGE). Next, 3' adapters were added, and 36–44 nt RNAs

enriched, followed by ligation of 5' adapters to RNA. Ligation products were subjected to reverse transcription via PCR amplification. PCR products 140–160 bp in size were enriched to generate a cDNA library and sequenced using Illumina HiSeq 2500 (Gene Denovo Biotechnology Co., Guangzhou, China).

The expression trends of DEGs were analyzed using the OmicShare Bioinformatics Cloud Platform (<https://www.omicshare.com/tools/>). The top of the graph shows the ID of the trend and the numbers within the graph represent the number of genes in the trend. Blocks with color ( $P < 0.05$ ) represent significantly enriched trends, with similar trend blocks of the same color, while blocks without color represent nonsignificantly enriched trends.

### Bioinformatic analysis

The phylogenetic relationship was analyzed by a phylogenetic tree, which was constructed by MEGA 7.0 with the Neighbor-Joining method and 1,000 bootstrap replicates. Multiple sequence alignment is performed at multalin (<http://multalin.toulouse.inra.fr/multalin/>).

### Reverse transcription-quantitative PCR analysis

Expression levels of genes were analyzed via reverse transcription-quantitative PCR (RT-qPCR) with the Light

Cycler 480 SYBR GREEN I Master mix (Roche, San Francisco, USA) according to the manufacturer's instructions. Gene expression levels were calculated using the  $2^{-\Delta\Delta CT}$  algorithm and normalized to *SITubulin* (Soylc08g006890). All analyses were repeated 3 times using 3 biological replicates. Statistical and correlation analyses were performed with SPSS 22.0 for Windows NT. The primers used are listed in Supplemental Table S4.

### Transient overexpression and VIGS in tomato fruit

The gene sequences of *WRKY71* cloned from cDNA of 'Micro Tom' tomatoes were used for experiments. PCR amplification was conducted using TransStart Fast *Pfu* DNA polymerase (AP221-01, Transgen, Beijing, China) and the primers listed in Supplemental Table S5. *WRKY71* was inserted between the *EcoRI* and *XhoI* restriction sites of the pSAK277 vector. Cultivation of the *Agrobacterium tumefaciens* strain GV3101 containing the appropriate vectors and transfection of tomato fruits was performed as described by Yao et al. (2017).

TRV-based vectors, pTRV1 and pTRV2, were used for VIGS. The *CAS1* and *WRKY71* fragments (400 bp in length corresponding to 1–400 nt) on CDS were amplified and inserted between *BamHI* and *EcoRI* sites of the pTRV2 vector to generate recombinant pTRV2-*CAS1* and pTRV2-*WRKY71*, respectively. Cultivation of the *A. tumefaciens* strain GV3101 containing the appropriate vectors and transfection of tomato fruits was performed as described by Fantini and Giuliano (2016).

### Plasmid construction and generation of transgenic tomato plants

The CDSs of *WRKY71* and *CAS1* were retrieved from the NCBI database. The gRNA target primers were generated based on the CRISPR direct platform (<http://crispr.dbcls.jp>), with 2 pairs designed per case (Supplemental Table S6). CRISPR/Cas9 vectors contained 2 gRNA targets of *WRKY71* and *CAS1*, respectively (Ma 2015). The primers used for the amplification of the fragment are presented in Supplemental Table S5. The primers for *CAS1* and *WRKY71* overexpression were designed using Primer Premier 5 (Supplemental Table S5). The CDSs of *CAS1* and *WRKY71* were inserted between *BamHI* and *XhoI* sites of the pBI121 vector. Plasmids were introduced into the *A. tumefaciens* strain EHA105, and tomato transformation was performed as described by Chen et al. (2003).

*CAS1* and *WRKY71* mutant tomato plants were identified by decoding sequencing results using DSDecodeM (<http://www.ygliulab.club/dsdecode>) and expression analysis performed using RT-qPCR. *CAS1*- and *WRKY71*-overexpressing transgene plants were identified via western blot and RT-qPCR analysis. Antigen affinity-purified antibodies of *WRKY71* and *CAS1* were produced using the specific peptides "CQVKKRVERSYQDP" and "NDAEKKGLITPGKTC,"

respectively, as antigens for immunization of rabbits (Genscript Biotech Corporation, Nanjing, China).

### Carotenoid, chlorophyll, and lycopene determination

Carotenoid and chlorophyll contents were determined according to the method of Nagata and Yamashita (1992). Briefly, 1.0 g tomato fruit powder was extracted with acetone and hexane (2:3 volume), and optical densities of the supernatant at 663, 645, 505, and 453 nm were measured using spectrophotometry. The contents of chlorophyll a, b, lycopene, and carotenoid were calculated from the equations as follows: chlorophyll a ( $\text{mg } 100 \text{ ml}^{-1}$ ) =  $0.999A_{663} - 0.0989A_{645}$ , chlorophyll b ( $\text{mg } 100 \text{ ml}^{-1}$ ) =  $-0.328A_{663} + 1.77A_{645}$ , lycopene ( $\text{mg } 100 \text{ ml}^{-1}$ ) =  $-0.0458A_{663} + 0.204A_{645} + 0.372A_{505} - 0.0806A_{453}$ , carotenoid ( $\text{mg } 100 \text{ ml}^{-1}$ ) =  $0.216A_{663} - 1.22A_{645} - 0.304A_{505} + 0.452A_{453}$ .

### Dual-luciferase reporter system assay

For the dual-luciferase reporter assay, the upstream promoter sequence of *CAS1* (2 kb) was cloned and inserted between *HindIII* and *BamHI* sites of pGreen II 0800-LUC vector using the primers listed in Supplemental Table S5. Recombinant plasmid was transformed into the *Agrobacterium* strain GV3101 (PM90) with the pSoup helper plasmid. The TFs *ERF109*, *ERFTINY*, *NAC73*, *NAC1*, *NAC35*, *WRKY2D*, *WRKY71*, *WRKY51*, and *WRKY65* were inserted into pSAK277 vector, mixed with promoter sequences at a ratio of 1:9 (v/v), and injected into *N. benthamiana* leaves for transient co-transformation expression analysis (Yoo et al. 2007). Five upstream activation sequences (UAS) were inserted into the vector with LUC reporter activity and the *WRKY71* fragment inserted into another vector containing 4 herpes simplex virus protein CDSs (VP16), followed by co-transfection into *N. benthamiana* leaves. The ratio of transactivation activities of firefly luciferase and renilla luciferase was determined with the Dual-Luciferase Reporter Assay System according to the manufacturer's instructions (E1910; Promega, Madison, USA).

### Luciferase complementation assay

Firefly luciferase complementation assays were performed according to the method of Chen et al. (2008). Gene sequences were amplified and cloned into pCAMBIA1300-nLuc (no stop codon) and pCAMBIA1300-cLuc vector. The primer sequences are listed in Supplemental Table S5. Recombinant vectors were subsequently transformed into *Agrobacterium* GV3101 and cultured in a selection medium containing kanamycin. At an OD<sub>600</sub> value of 0.6, *A. tumefaciens* was mixed at a ratio of 1:1 (v/v) and infiltrated into *N. benthamiana* leaves (Chen et al. 2008). Leaf disks (2 cm in diameter) were punched adjacent to the infiltration site, and the firefly luciferase activity was determined using a Steady-Glo Luciferase Assay System (E2510, Promega, Madison, USA).



### Y1H assay

The sequence containing the WRKY binding site of the CAS1 promoter was amplified and inserted into the pAbAi bait vector containing *Hind* III and *Xho* I, while WRKY71 was cloned into the pGADT7 prey vector containing *Eco*R I and *Xho* I. After filtering by 0.45 µm membrane, self-activation of bait vectors was tested on SD/-ura + AbA<sup>100</sup>, SD/-ura + AbA<sup>200</sup>, and SD/-ura + AbA<sup>400</sup> plates, while the prey vector, pGADT7-WRKY71, was tested on SD/-Leu plates. The promoter segment baits were fused to the prey vector, introduced into the Y1HGold yeast strain, and tested on SD/-ura + AbA plates at 30 °C for 3 d (Li et al. 2020a, 2020b). The primer sequences used for vector construction are listed (Supplemental Table S5).

### Y2H assay

Full-length WRKY71 sequences were cloned into pGBKT7 containing *Nde* I and *Pst* I for the transactivation activity assay. Full-length BRG3 was cloned into pGADT7 containing *Eco*R I and *Xho* I and co-transformed in pairs into yeast strain AH109 cells using the LiAC-PEG method. The primers used are presented in Supplemental Table S5. Selection of co-transformed colonies was performed on SD(-Leu/-Trp) growth medium. Protein interactions were examined via growth on SD(-Leu/-Trp/-His/-Ade) screening medium. Co-transformations of pGADT7-T and pGBKT7-53/pGBKT7-Lam were used as the positive and negative control, respectively (Li et al. 2020a, 2020b).

### Prokaryotic expression vector construction and recombinant protein expression

The full-length BRG3 sequence was cloned between the *Bam*HI and *Hind*III restriction sites of pMAL-c2x with MBP tag and full-length WRKY71 cloned between the *Eco*RI and *Xho*I restriction sites of pGEX-4T-1 with GST tag. The primers used for cloning are presented in Supplemental Table S5. Recombinant plasmids were transformed into DE3 for protein expression. Transformants carrying pMAL-BRG3 and pGEX-WRKY71 were grown overnight at 37 °C in LB containing ampicillin. The culture was subsequently transferred to 100 ml fresh LB containing ampicillin at 37 °C until an OD<sub>600</sub> value of 0.6–0.8 was reached. Isopropyl-β-D-thiogalactopyranoside (IPTG) was added at a final concentration of 1 mM and proteins were induced overnight by incubation at 16 °C and 120 rpm (Clos and Brandau 1994). BRG3 protein with MBP tag was purified using Amylose Resin (E8021V, NEB, England), and WRKY71 protein with GST tag was purified using Glutathione Sepharose 4B (10305042; Cytiva, Sweden).

### Biotin switch assay of protein persulfidation

Persulfidation was detected with the biotin conversion method using purified BRG3 protein treated with H<sub>2</sub>S as described by Ma et al. (2021). The protocol was as follows: BRG3 treated with H<sub>2</sub>S and acetone-concentrated was dissolved in 100 µl HEN (250 mM Hepes-NaOH (pH 7.7), 1 mM EDTA

and 0.1 mM neocuproine) at different concentrations. NaHS and dithiothreitol were incubated at 4 °C for 30 min. The protein was re-precipitated with acetone and re-dissolved in 100 µl HEN buffer. Following the addition of methyl thiosulfonate blocking solution (400 µl), the mixture was allowed to react overnight at 4 °C. Next, the acetone precipitate was dissolved in 100 µl of 1% SDS HEN buffer and 30 µl biotin-N-[6-(biotinamino) hexyl]-3'-(2'-pyridyl disulfide) base-propionamide solution and the mixture reacted at 25 °C for 3 h. Subsequently, the protein was precipitated with acetone, dissolved in 80 µl HEN buffer using biotin and HRP-conjugated antibodies, and a High-sig ECL substrate was used for color development. LC-MS/MS-based peptide identification was performed as described by Rauh (2012) and confirmed using the UniProtKB database (<https://www.uniprot.org/help/uniprotkb/>).

### In vitro ubiquitination assay

The *in vitro* ubiquitination assay was performed according to the method of Abramovitch et al. (2006) with slight modifications. The ubiquitination reaction mixture (30 µl) comprising ubiquitination buffer (50 mM Tris HCl, pH 7.5, 2 mM ATP, 5 mM MgCl<sub>2</sub>, 30 mM phosphocreatine [Sigma-Aldrich, Saint Louis, USA]), 50 ng µl<sup>-1</sup> creatine phosphokinase (Sigma-Aldrich), recombinant E1, E2, and ubiquitin (a small molecule protein) (Unibi Biotech, Changchun, Jilin, China) was incubated at 37 °C for 12 h. Proteins were separated using 7.5% SDS-PAGE and identified via western blot with anti-Ub antibody (dilution is 1:5,000) and anti-GST antibody (dilution is 1:5,000) (Beyotime Biotechnology, Shanghai, China).

### Accession numbers

Sequence data from this article can be found in the National Center for Biotechnology Information (NCBI, <https://www.ncbi.nlm.nih.gov/>) or the Phytozome (<https://phytozome-next.jgi.doe.gov/>) under numbers: WRKY71 (LOC101264783), CAS1 (LOC101253940), BRG3 (LOC101267168), RNF-144b (LOC101265447), ERFTINY (LOC101259323), ERF109 (LOC101246484), NAC73 (LOC101264084), NAC35 (LOC101267943), NAC1 (Solyc04g009440.2), WRKY2D (Solyc02g093050), WRKY51 (LOC101258361), WRKY65 (LOC101267623), SGR1 (Solyc08g080090), PPH (Solyc01g088090), PAO (Solyc11g066440), NYC1 (Solyc07g024000), PSY1 (Solyc03g031860), PDS (Solyc03g123760), ZDS (Solyc01g097810).

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### Supplemental data

The following materials are available in the online version of this article.

**Supplemental Figure S1.** Evolutionary relationships and comparison of protein structures of WRKY TF from different species.

**Supplemental Figure S2.** Functional analysis of WRKY71 in tomato fruit via the transient overexpression assay.

**Supplemental Figure S3.** Functional analysis of WRKY71 in tomato fruit via the virus-induced gene silencing (VIGS) assay.

**Supplemental Figure S4.** Identification of CRISPR/Cas9-WRKY71 tomato fruit (*wrky71*) and stable WRKY71 transgene tomato (WRKY71-OE).

**Supplemental Figure S5.** Analysis of CAS1 in tomato fruit via the virus-induced gene silencing (VIGS) assay.

**Supplemental Figure S6.** Identification of CRISPR/Cas9-CAS1 tomato fruit (*cas1*) and stable CAS1 transgene tomato (CAS1-OE).

**Supplemental Figure S7.** Detection of persulfidation signal in CAS1.

**Supplemental Table S1.** Transcriptome data filtering and quality assessment.

**Supplemental Table S2.** Correlation analysis of *ERF109*, *NAC35*, *NAC1*, *WRKY2D*, *NAC73*, *WRKY51*, *WRKY71*, *WRKY65*, and *CAS1* expressions.

**Supplemental Table S3.** Identification of the persulfidation peptide by LC-MS/MS.

**Supplemental Table S4.** Primers used in this study for RT-qPCR analysis.

**Supplemental Table S5.** Primers used in this study for constructing experiment-related vectors.

**Supplemental Table S6.** Primers used for PCR amplification of fragments containing the target sites of gRNA of CRISPR/Cas9-WRKY71 and CRISPR/Cas9-CAS1.

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